Indian Journal of Chemistry
Vol. 31A, December 1992, pp. 903-908

Microstructure and glass transition temperature determination of styrene-methyl acrylate copolymers by ^{13}C-NMR spectroscopy

A S Brar* & Sunita

Department of Chemistry, Indian Institute of Technology, Hauz Khas, New Delhi 110016, India

Received 10 February 1992; revised and accepted 27 April 1992

Styrene-methyl acrylate (S/M) copolymers of different monomer compositions have been prepared by bulk polymerization using benzoyl peroxide as an initiator. Copolymer composition is determined from NMR spectrum and comonomer reactivity ratios have been determined by the method of Kelen-Tudos. The copolymerization parameters such as terminal, penultimate reactivity ratios, diad concentrations, conditional probabilities, number-average sequence lengths and run number in the copolymers have been calculated from the observed triad sequence distribution determined from ^{13}C[^{1}H]-NMR spectra. The observed triad sequence concentrations as determined from ^{13}C[^{1}H]-NMR spectroscopy are in good agreement with those calculated from reactivity ratios. Glass transition temperature (T_g) values of various copolymers calculated from NMR spectra (using Barton's equation) give good agreement with the values obtained from DSC.

In our earlier papers we have reported the microstructure of acrylonitrile-methyl methacrylate[1], styrene-methyl methacrylate[2], acrylonitrile-ethyl methacrylate[3] and acrylic acid-vinyl acetate[4] copolymers using ^{13}C-NMR spectroscopy. This study is a continuation of earlier work in which microstructure parameters, e.g., terminal and penultimate reactivity ratios, number-average sequence lengths, run number, block character and T_g of styrene-methyl acrylate copolymers have been calculated from triad and diad concentrations obtained from ^{13}C[^{1}H]-NMR spectra. The effects of copolymer composition and monomer sequence distribution on T_g has been investigated for some copolymers[5-8]. Koinuma *et al.*[9] have theoretically calculated the tacticity and sequence dependence of ^{13}C-NMR spectra of styrene-methyl acrylate copolymer. De Vries *et al.*[10] have reported the intramolecular structure (triad distribution and tacticity parameters) of S/M copolymers obtained by low conversion solution polymerization and high conversion by mean of both solution and emulsion polymerization. They have also used ^{1}H-NMR to determine the fractions of various S- and M-centered triads but we have used only ^{13}C-NMR spectra to calculate the fractions of various triads because of the poor resolution of ^{1}H-NMR spectra.

Materials and Methods

Polymerization

Monomers were distilled under reduced pressure and stored below 5°C. Benzoyl peroxide (0.5% by weight) was added to a mixture of monomers and polymerization was carried out at 70°C. The conversion was kept low (5-10%) and copolymers were precipitated in excess of methanol after 40 min. Several copolymers having various compositions were prepared similarly.

NMR Analyses

^{1}H-NMR and ^{13}C[^{1}H]-NMR spectra were recorded on the Bruker W H 270 NMR spectrometer operating at 270 MHz and 67.5 MHz respectively. The detailed conditions of the operations are as follows: temperature of the probe, 25°C; reference, centre peak of $CDCl_3$ assigned as 77.0 ppm; spectral width, 13500 Hz; pulse repetition time, 5 sec; 16 K data points for fourier transform. The details of recording the spectra and Lorentzian shape curve fitting have been described elsewhere[2,3].

Molecular weight determination

The molecular weight distribution of styrene-methyl acrylate copolymers was determined on Water's GPC instrument at 25°C using a 5 μ-PL-gel mixed column. Chloroform was used as the mobile phase and a refractive index detector was used. Calibration was done with narrow molecular weight distributed poly(methyl methacrylate) (PMMA) as reference standard.

Results and Discussion

Reactivity ratios determination

The composition of styrene-methyl acrylate (S/M) copolymers prepared by bulk polymeriza-

tion was found from ^{1}H-NMR spectra of the copolymers by comparing the intensities of the $-OCH_3$ (M unit) and $-C_6H_5$ (S unit) proton resonances. The styrene mole fraction (F_S) in the copolymers can readily be determined by using the relationship:

$$F_S = \frac{3I(-C_6H_5)}{3I(-C_6H_5)+5I(-OCH_3)} \quad \ldots (1)$$

where $I(-OCH_3)$ and $I(-C_6H_5)$ represent the intensities of $-OCH_3$ and $-C_6H_5$ proton resonances respectively. To ascertain normal copolymer kinetic behaviour, a plot of theoretical composition of the copolymer obtained from copolymer equation by using terminal model reactivity ratios ($r_S = 1.20$, $r_M = 0.18$) along with experimental data was drawn (Fig. 1). Both these values are in good agreement with each other and the concave shape of the curve indicates that the monomer distribution is statistical in nature and in no case homopolymer formation is expected at low conversion. The copolymer composition data were used to calculate the terminal model reactivity ratios by the method of Kelen-Tudos[11]. The values of reactivity ratios obtained from Kelen-Tudos plot are $r_S = 1.20$ and $r_M = 0.18$. The value of the product of r_S. r_M is about 0.22, which indicates that S/M comonomer pair polymerizes neither in random nor in alternate manner. The reported values[10] of the reactivity ratios are $r_S = 0.73$ and $r_M = 0.19$. In the reported study the copolymers were prepared by the solution and emulsion polymerization through free radical initiation, while in the present case copolymers were prepared by bulk polymerization as explained earlier. This indicates that the values of reactivity ratios depend on the method of polymerization.

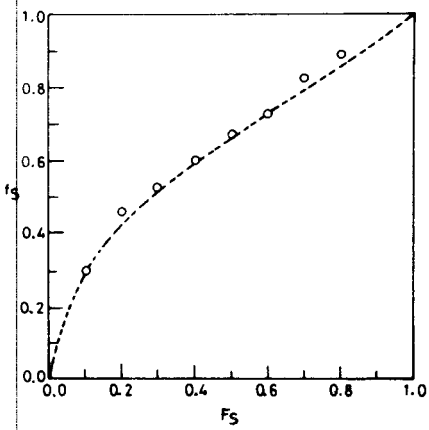


Fig. 1 – The theoretical copolymer composition curve (– – –) obtained from copolymer composition equation by using terminal model reactivity ratios ($r_S = 1.20$, $r_M = 0.18$) along with the experimental points (0).

Molecular weight distribution

The PMMA equivalent number average ($\bar{M}_n$), weight average ($\bar{M}_W$), Z average ($\bar{M}_Z$) and Z+1 average ($\bar{M}_{Z+1}$) molecular weights and the polydispersity of the S/M copolymers determined from GPC measurements are given in Table 1. The $\bar{M}_n$ varies from 0.40 to 0.76×10^5, $\bar{M}_W$ varies from 0.69 to 1.26×10^5, $\bar{M}_Z$ varies from 1.19 to 2.04×10^5, $\bar{M}_{Z+1}$ varies from 1.86 to 3.06×10^5 and $\bar{M}_V$ varies from 0.69 to 1.26×10^5. The polydispersity value of all the copolymers is almost of the same order (≈ 1.7), so the heterogeneity of the copolymers and the molecular weight distribution is of the same order.

^{1}H-NMR studies

In the case of S/M copolymer, the $-(OCH_3)$ protons appeared around 3.6 ppm. The broad signal from 2.3 to 2.0 ppm can be attributed to $(-CH)$ proton of both styrene and methyl acrylate monomers. The other broad signal from 1.7 to 1.3 ppm can be assigned to $(-CH_2)$ protons of both monomers. The aromatic protons in all the copolymers appeared as a doublet, one due to *ortho* (6.6-6.8 ppm) and another due to *meta* and *para* (7.1-7.8 ppm) protons.

^{13}C-NMR studies

Various resonance signals in the $^{13}C[^1H]$-NMR spectrum of S/M copolymer have been assigned by comparing the spectrum of the copolymer with those of homopolymers. In the case of polystyrene (PS), a sharp singlet centered around 40.8 ppm can be assigned to $(-CH)_S$. The methylene $(-CH_2)_S$ carbon shows a multiplet in the range 43.0-46.0 ppm, which shows sensitivity toward tacticity. In the multiplet, the peak around 43.4 ppm can be assigned to isotactic polystyrene, that at 44.8 to heterotactic and that at 45.7 ppm can be assigned to syndiotactic polystyrene. The aromatic carbons in polystyrene show splitting ac-

Table 1 – Study of molecular weight distribution of styrene-methyl acrylate (S/M) copolymers

S.No.	S. mol. fraction (in feed)	$\bar{M}_n$ [10^{-5}]	$\bar{M}_W$ [10^{-5}]	$\bar{M}_Z$ [10^{-5}]	$\bar{M}_V$ [10^{-5}]	$\bar{M}_{Z+1}$ [10^{-5}]	Polydispersity ($\bar{M}_W/\bar{M}_n$)
1.	0.7	0.74	1.24	2.04	1.24	3.06	1.67
2.	0.6	0.60	0.97	1.54	0.97	2.24	1.63
3.	0.5	0.76	1.26	1.98	1.26	2.79	1.65
4.	0.4	0.59	1.04	1.78	1.04	2.67	1.76
5.	0.3	0.40	0.69	1.19	0.69	1.86	1.75
6.	0.2	0.60	1.03	1.71	1.03	2.71	1.71

cording to their environments. A highly complex aromatic ring carbon resonance pattern is exhibited by the quaternary carbon ($-\dot{C}-$) around 145.0-146.1 ppm, while no splitting was observed for either the *ortho*, *meta* or *para* carbon, which appeared around 128.6, 127.2 and 126.0 ppm respectively. The multiplicity of quaternary carbon indicates its sensitivity towards configurational sequence and a peak around 146.0 ppm can be assigned to isotactic polystyrene, one around 145.5 ppm can be assigned to heterotactic and the peak around 145.2 ppm can be assigned to syndiotactic polystyrene[12].

In the case of PMA, resonance signals around 174.7, 51.2, 42.1 and 34.5 ppm can be assigned to $(>C=0)_M$, $(OCH_3)_M$, $(-CH)_M$ and $(-CH_2)_M$ carbons respectively. $^{13}C[^1H]$-NMR spectrum of S/M copolymer (S = 53.0 mole per cent in copolymer) recorded in $CDCl_3$ at room temperature along with assignments and the positions of various functional groups is shown in Fig. 2. The quaternary carbon of S unit appeared as a well resolved multiplet at 143.0-145.8 ppm, showing its sensitivity towards different monomer placements. The carbonyl carbon resonance signal in the S/M copolymer appeared as a multiplet around 176.5-174.5 ppm, which indicates that the splitting of carbonyl signal is due to its sensitivity toward different compositional sequences. In the case of S/M copolymer, the positions of various functional groups of S and M units shift as compared to the positions in the homopolymers, which may be due to the change in the nature of adjacent monomeric units in copolymer which changes the chemical shift of S and M centered triad.

The carbonyl ($>C=O$) carbon expansion of S/M copolymer (S = 53.0 mole per cent in copolymer) is shown in Fig. 2. Free radical PMA shows a singlet around 174.7 ppm. As the per cent of styrene in the copolymer increases, signals characteristic of homopolymer decrease, whereas set of signals centered around 175.4 ppm start appearing. These signals, with a further increase in the S content, increase to a maximum value and then decrease, whereas a third new set of resonance signal appears around 176.2 ppm. Three sets of signals whose intensities change with the copolymer composition can be assigned to carbonyl carbon centered in SMS, SMM (MMS), MMM triad sequences from low to high field. In the case of M-centered triad, introduction of S unit in MMM triad causes the downfield shift in the position of SMM (by ≈ 0.8 ppm) and SMS (by ≈ 0.70 ppm). Concentrations of various M-centered triads can be calculated from the relative areas of respective signals. Table 2 contains the compositional information concerning various M-centered triads along with calculated values obtained from Alfrey-Mayo model[13] (first order

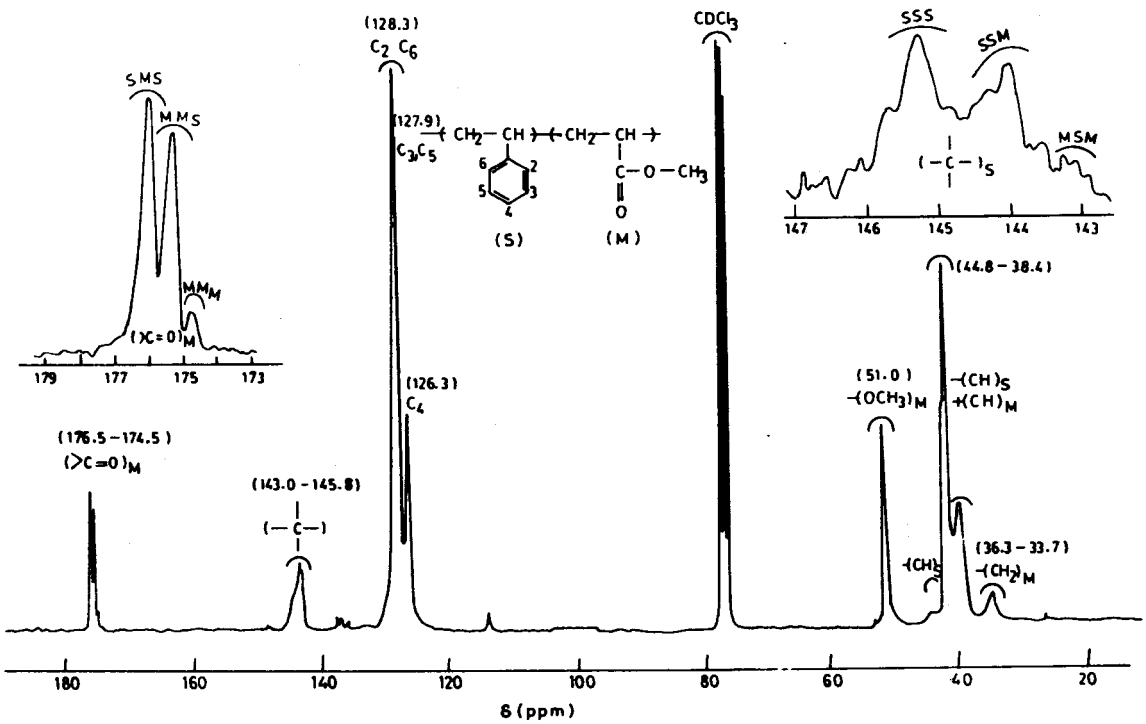


Fig. 2 – $^{13}C[^1H]$-NMR spectrum of styrene-methyl acrylate copolymer (S = 53.0 mole per cent in copolymer) in $CDCl_3$ at room temperature along with the expanded carbonyl $(>C=O)_M$ and quaternary $(-\dot{C}-)_S$ carbon resonance signals.

Table 2 – Calculated and observed fractions of S- and M-centered triad in styrene-methyl acrylate (S/M) copolymers

S.No.	S. mol. fraction (in feed)	Triad	Triad Obs.	Concentrations Calc. from		Penultimate reactivity ratios
				Alfrey-Mayo model	Harwood Pen. model	
1.	0.7	SSS	0.54	0.52	0.53	$r_{SS} = 1.01$
		SSM	0.39	0.44	0.40	$r_{MS} = 2.35$
		MSM	0.07	0.04	0.07	$r_{MM} = 0.00$
		MMM	0.00	0.00	0.00	$r_{SM} = 0.17$
		MMS	0.14	0.13	0.13	
		SMS	0.86	0.87	0.87	
2.	0.6	SSS	0.41	0.38	0.40	$r_{SS} = 1.15$
		SSM	0.46	0.44	0.47	$r_{MS} = 0.82$
		MSM	0.13	0.18	0.13	$r_{MM} = 0.15$
		MMM	0.01	0.01	0.01	$r_{SM} = 0.20$
		MMS	0.19	0.21	0.19	
		SMS	0.80	0.78	0.80	
3.	0.5	SSS	0.30	0.30	0.29	$r_{SS} = 1.33$
		SSM	0.49	0.45	0.50	$r_{MS} = 0.90$
		MSM	0.21	0.25	0.21	$r_{MM} = 0.20$
		MMM	0.02	0.03	0.02	$r_{SM} = 0.22$
		MMS	0.26	0.30	0.26	
		SMS	0.72	0.67	0.72	
4.	0.4	SSS	0.20	0.22	0.19	$r_{SS} = 1.43$
		SSM	0.49	0.46	0.50	$r_{MS} = 1.08$
		MSM	0.31	0.32	0.31	$r_{MM} = 0.20$
		MMM	0.05	0.05	0.04	$r_{SM} = 0.18$
		MMS	0.33	0.33	0.34	
		SMS	0.62	0.62	0.62	
5.	0.3	SSS	0.11	0.08	0.11	$r_{SS} = 0.85$
		SSM	0.45	0.44	0.45	$r_{MS} = 1.07$
		MSM	0.44	0.48	0.44	$r_{MM} = 0.13$
		MMM	0.08	0.06	0.07	$r_{SM} = 0.17$
		MMS	0.42	0.41	0.42	
		SMS	0.50	0.53	0.51	
6.	0.2	SSS	0.05	0.05	0.05	$r_{SS} = 1.18$
		SSM	0.36	0.34	0.36	$r_{MS} = 1.11$
		MSM	0.59	0.61	0.59	$r_{MM} = 0.17$
		MMM	0.17	0.16	0.15	$r_{SM} = 0.17$
		MMS	0.49	0.48	0.50	
		SMS	0.34	0.36	0.35	

Markov terminal model) using reactivity ratios ($r_S = 1.20$, $r_M = 0.18$). The relation between intramolecular structure (fractions of S and M centered triad) are given below:

$$F_{MMM} = [1 - P_{(S/M)}]^2 \quad \ldots (2a)$$

$$F_{MMS} = F_{SMM} = 2[P_{(S/M)}].[1 - P_{(S/M)}] \quad \ldots (2b)$$

$$F_{SMS} = [P_{(S/M)}]^2 \quad \ldots (2c)$$

$$F_{SSS} = [1 - P_{(M/S)}]^2 \quad \ldots (2d)$$

$$F_{SSM} = F_{MSS} = 2[P_{(M/S)}].[1 - P_{(M/S)}] \quad \ldots (2e)$$

$$F_{MSM} = [P_{(M/S)}]^2 \quad \ldots (2f)$$

where $P_{(S/M)} = \left(\frac{1}{1 + r_M/q}\right)$ and $P_{(M/S)} = \left(\frac{1}{1 + r_S \cdot q}\right)$

$q = [S]/[M]$ is the instantaneous feed ratio. F represents the fractions of triad normalized to unity. Penultimate reactivity ratios were evaluated (Table 2) from experimentally obtained triad distributions from carbonyl carbon resonance of $^{13}C[^1H]$-NMR spectra by using the following equations:

$$r_{MM} = \frac{2\,[MMM]}{[MMS + SMM]} \cdot [S_f]/[M_f] \quad \ldots (3a)$$

$$r_{SM} = \frac{[SMM + MMS]}{2[SMS]} \cdot [S_f]/[M_f] \quad \ldots (3b)$$

The values of r_{MM} and r_{SM} are listed in Table 2. The values of penultimate reactivity ratios vary with copolymer composition.

Similar compositional information regarding S-centered triads can be made using the $(-\underset{t}{C}-)$ carbon resonance region. The quaternary carbon $(-\underset{t}{C}-)$ resonance expansion of S/M copolymer (S = 53.0 mole per cent in copolymer) is shown in Fig. 2. A multiplet around 143.0-145.8 ppm shows the splitting into three envelopes. The chemical shift difference within the resonance signals and splitting pattern shows its sensitivity towards monomer sequence and cotacticity in the quaternary carbon resonance. As the concentration of S unit in the copolymer increases the most downfield signal of the quaternary carbon centered around 145.2 ppm increases, while the upfield signal centered around 143.2 ppm decreases. On the basis of compositional variation in the intensities of S-centered triad, the most downfield resonance signal may be assigned to SSS triad and the most upfield due to MSM triad. The addition of methyl acrylate to SSS centered triad causes the upfield shift in the chemical shift of SSM (by ≈ 0.9 ppm) and MSM (by ≈ 1.1 ppm). S-Centered triad distributions along with those calculated using Alfrey-Mayo model are given in Table 2. On the basis of the triad resonance assignments made as described earlier, penultimate reactivity ratios were evaluated for S/M copolymer from the triad fractions determined from qua-

ternary carbon resonance of $^{13}C[^1H]$-NMR spectra by using the following equations:

$$r_{SS} = \frac{2[SSS]}{[SSM + MSS]} \cdot [M_f]/[S_f] \quad \ldots (3c)$$

$$r_{MS} = \frac{[SSM + MSS]}{2[MSM]} \cdot [M_f]/[S_f] \quad \ldots (3d)$$

The values of r_{SS} and r_{MS} are listed for individual copolymers in Table 2. Using the average values of penultimate reactivity ratios, the triad sequence distribution as obtained by using Harwood program[14] is given in the Table 2.

From the results of composition of various S- and M-centered triads, the conditional probabilities P_{SM} and P_{MS} have been calculated using the equations:

$$P_{SM} = \frac{[MSM] + [MSS]/2}{[MSM] + [MSS] + [SSS]} \quad \ldots (4a)$$

$$P_{MS} = \frac{[SMS] + [MMS]/2}{[MMM] + [MMS] + [SMS]} \quad \ldots (4b)$$

where P_{SM} is the probability that S-M unit comes about as a result of S growing chain end adding M and P_{MS} is the probability that M-S unit comes about as a result of M growing chain end adding S. The value of P_{SM} increases from 0.22 to 0.78 with the increase in the content of methyl acrylate in the copolymers while the value of P_{MS} decreases from 0.93 to 0.60 with the decrease in the concentration of styrene in the copolymers. The number average sequence lengths ($\bar{N}_s$ and $\bar{N}_M$), being the reciprocal of the conditional probabilities, were also calculated. The value of $\bar{N}_S$ decreases from 4.46 to 1.28 with the decrease in the concentration of S in the copolymer, while the values of $\bar{N}_M$ increases from 1.07 to 1.67 with the decrease in the content of S in the copolymers.

The block character (η) is an adequate way to characterize the sequence distribution in the copolymers. It is given by the expression[15]:

$$\eta = \frac{[SM]}{2[S][M]} \quad \ldots (5)$$

where [SM] is the mole fractions of SM dyad and [S], [M] are the mole fractions of S and M monomers in the copolymers. The value $0 < \eta < 1$ reflects more block character and $1 < \eta < 2$ means more alternating tendency of the copolymer than that expected from the random distribution. The value of η for S/M copolymerization lies in the range 0.67 - 0.88 ($0 < \eta < 1$), which shows that S/M copolymer has more tendency toward block copolymerization.

The run number (R) of this copolymer is defined as the average number of the segments of either type per 100 monomer units. This may be written as[16]:

$$R = \frac{200}{\bar{N}_S + \bar{N}_M} \quad \ldots (7)$$

where $\bar{N}_S$ and $\bar{N}_M$ are the number average sequence lengths of S and M monomer units in the copolymers. In the case of S/M copolymer, the value of R lies in the range 36.17-71.68. All the copolymerization parameters are given in Table 3.

Barton[17] and Johnston[18] described the effect of sequence distribution of monomeric unit in the linear copolymers on its glass transition temperature (T_g) and considered the contribution of the diad. Barton equation showed the dependence on sequence as follows:

$$T_{g,P} = N_{SS} T_{g,SS} + N_{MM} T_{g,MM} + (N_{SM} + N_{MS}) T_{g,SM}. \quad \ldots (8)$$

$T_{g,P}$ is the T_g of a copolymer containing mole fractions (N) of various diads (SS, MM, SM, MS). $T_{g,SS}$ and $T_{g,MM}$ are the glass transition temperatures of polystyrene and polymethyl acrylate respectively, whereas $T_{g,SM}$ is the T_g of the alternating copolymer. Barton[17] has reported the value of T_{SM} for styrene-methyl acrylate copolymer as 58.3°C. This value seems to be logical looking from the view point of T_g of homopolymers. The values of T_g for polystyrene was taken as 100°C

Table 3 – Copolymerization parameters of S/M copolymer determined by ^{13}C-NMR spectroscopy

S.No.	S. mol. fraction (in feed)	P_{SM}	P_{MS}	$\bar{N}_S$	$\bar{N}_M$	$[\eta]$	R
1.	0.7	0.22	0.93	4.46	1.07	0.67	36.17
2.	0.6	0.40	0.88	2.50	1.13	0.73	55.10
3.	0.5	0.47	0.82	2.11	1.22	0.67	60.06
4.	0.4	0.50	0.78	2.00	1.27	0.69	61.16
5.	0.3	0.70	0.73	1.43	1.36	0.88	71.68
6.	0.2	0.78	0.60	1.28	1.67	0.78	67.80

Table 4 – Glass transition temperature (T_g) and diad compositions data for S/M copolymers

S.No.	S. mole. fraction (in feed)	Diad fractions	T_g (K)	
			Calc. from Barton eqn.	Obtained from DSC
1.	0.7	[SS] = 0.74 [MM] = 0.07 [MS] = 0.19	359	360
2.	0.6	[SS] = 0.60 [MM] = 0.12 [MS] = 0.28	350	349
3.	0.5	[SS] = 0.53 [MM] = 0.18 [MS] = 0.29	344	343
4.	0.4	[SS] = 0.45 [MM] = 0.22 [MS] = 0.33	339	–
5.	0.3	[SS] = 0.30 [MM] = 0.26 [MS] = 0.44	330	–
6.	0.2	[SS] = 0.22 [MM] = 0.40 [MS] = 0.38	320	–

and of polymethyl acrylate as 6°C. By using the Barton's equation (Eq. 8), the values of T_g were calculated for these copolymers. The values thus calculated were found to be in agreement with the values obtained from DSC. Table 4 contains the diad concentrations calculated from $^{13}C[^1H]$-NMR spectra along with the T_g data. The T_g's of the copolymers having less than 60% of styrene (in copolymer) are not reported here because these are very low.

Acknowledgement

The authors are thankful to Prof. C.L. Khetrapal, I.I.Sc. Bangalore, for recording ^{13}C-NMR spectra.

References

1 Kapur G S & Brar A S, *Polymer*, 32 (1991) 1112.
2 Brar A S & Kapur G S, *Polym J*, 20 (1988) 371.
3 Kapur G S & Brar A S, *J polym Sci polym Chem*, 29 (1991) 479.
4 Brar A S & Sunita, *Eur Polym J*, 27 (1991) 17.
5 Sanchez-Chaves M, Arranz F & Montes M, *Polymer*, 29 (1988) 2244.
6 Arranz F, Sanchez-Chaves M & Molinero A, *Macromol Chem*, 185 (1984) 2153.
7 Harris S H & Gilbert R D, *J polym Sci polym Chem Ed*, 20 (1982) 1653.
8 Schneider H A & Neto H N, *Polym Bull*, 9 (1983) 457.
9 Sato K, Koinuma H & Hirai H, *Macromol Chem*, 4 (1983) 821.
10 van Doremaele G H J, German A L, De Vries N K & van der veldon G P M, *Macromolecules*, 23 (1990) 4206.
11 Kelen T & Tudos F, *J Macromol Sci Chem*, A9 (1975) 1.
12 Randall J C, *Polymer sequence determination: Carbon-13 NMR method*, Chap. 6 (Academic Press, New York), 1977, 116.
13 Koenig J L, *Chemical microstructure of polymer chain*, Chap. 3 (Wiley Interscience, New York), 1980, 39.
14 Harwood H J, *J polym Sci Part C*, 25 (1968) 37.
15 Ito K & Yamashita Y, *J polym Sci*, A3 (1965) 2165.
16 Allcock H R & Lampe F W, *Contemporary polymer chemistry* Chap. 12/II Ed. (Englewood Cliffs, New Jersey), 1990, 116.
17 Barton J M, *J polym Sci Part C*, 30 (1970) 573.
18 Johnston N W, *J Macromol Sci rev Macromol Chem*, 14 (1976) 215.